Day 3 — Why It Works — The Algebra

Module: ×11 Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive (10a+b)×11 = 100a + 10(a+b) + b and explain.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: Anurūpyena; lit. ““proportionately” — sūtra used for ×12, ×13 extensions”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to *“What’s one thing you remember about yesterday’s lesson?”*
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Write any 2-digit number as 10a + b.** Examples: 23 = 10·2 + 3 (a=2, b=3); 87 = 10·8 + 7.

2. **Distribute multiplication.**

   ```
   (10a + b) × 11 = (10a + b)(10 + 1)
                  = (10a + b)·10 + (10a + b)·1
                  = 100a + 10b + 10a + b
                  = 100a + 10(a + b) + b
   ```

3. **Read the final line in plain language.**

   - 100a → hundreds digit is *a*
   - 10(a+b) → tens digit is *a*+*b* (the digit sum!)
   - +b → units digit is *b*

   This IS the trick. The algebra spells out the procedure.

4. **Verify on 23.** 100·2 + 10·(2+3) + 3 = 200 + 50 + 3 = 253. ✓

5. **Connection to the carry case.** When $a+b \geq 10$, $10(a+b)$ becomes $10(a+b) = 100 + 10\cdot(a+b-10)$, pushing the overflow into the hundreds place. The algebra automatically handles the carry.

6. **Independent task.** Each student writes the proof in their own words for a homework explainer to a younger student.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: *“What’s one thing you learned today?”*
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).